ONE PERSON IN GROUP-> CLICK USE TEMPLATE

WHEN THE COPIED TEMPLATE OF THESE SLIDES OPENS IN A NEW TAB:

- ADD YOUR Group NAME TO THE TITLE
- SHARE WITH GROUP MEMBERS (USE SHARE BUTTON)
- COPY THE TITLE AND LINK TO YOUR RECORDS (Digital folder/table of contents)
- Remember it is advisable to make a PERSONAL copy and link it when DONE

Dilutions

ESSENTIAL QUESTIONS

Remember, you should be able to thoroughly answer these AFTER learning through following resources.

- How can we calculate dilutions and concentrations we need in the lab?
- How can we safely prepare dilutions for our needs?

Instructions

Follow through the Engage to Apply resources in order .

- Once complete, choose from Extend resources as interested.
- Record your learning in your these slides- Remember to keep track of where to find these slides for studying!
- Complete the Essential Questions Response in the Apply Section .
- Try the Extend section and optional Apply practice

Hyperdoc Elements

Click for quick access to:

- Engage
- Explore
- Explain
- Apply
- Extend

Terminology - As you encounter terms new to you, define in your own words here, include examples (make them up!)

-

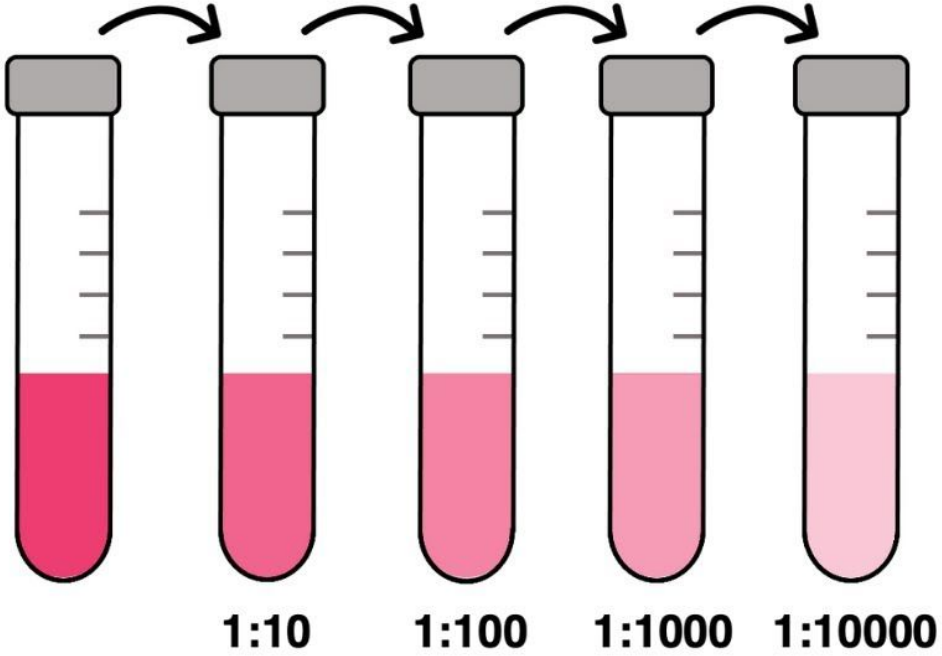
Part 1: ENGAGE

Explore - Part A: 5 minutes

Make two observations & one inference about the image on the right:

Observations:

Inference:



Part 2: EXPLORE

Explore - Preparing Solutions

Preparing solutions with known concentrations is an important skill in chemistry

- Prepared Solutions with known concentrations - **Standard Solution**
- Purchased Solutions from a Chemical Provider - **Stock Solution**
    - Both used for accurate stoichiometric calculations

There are two ways to prepare Standard Solutions:

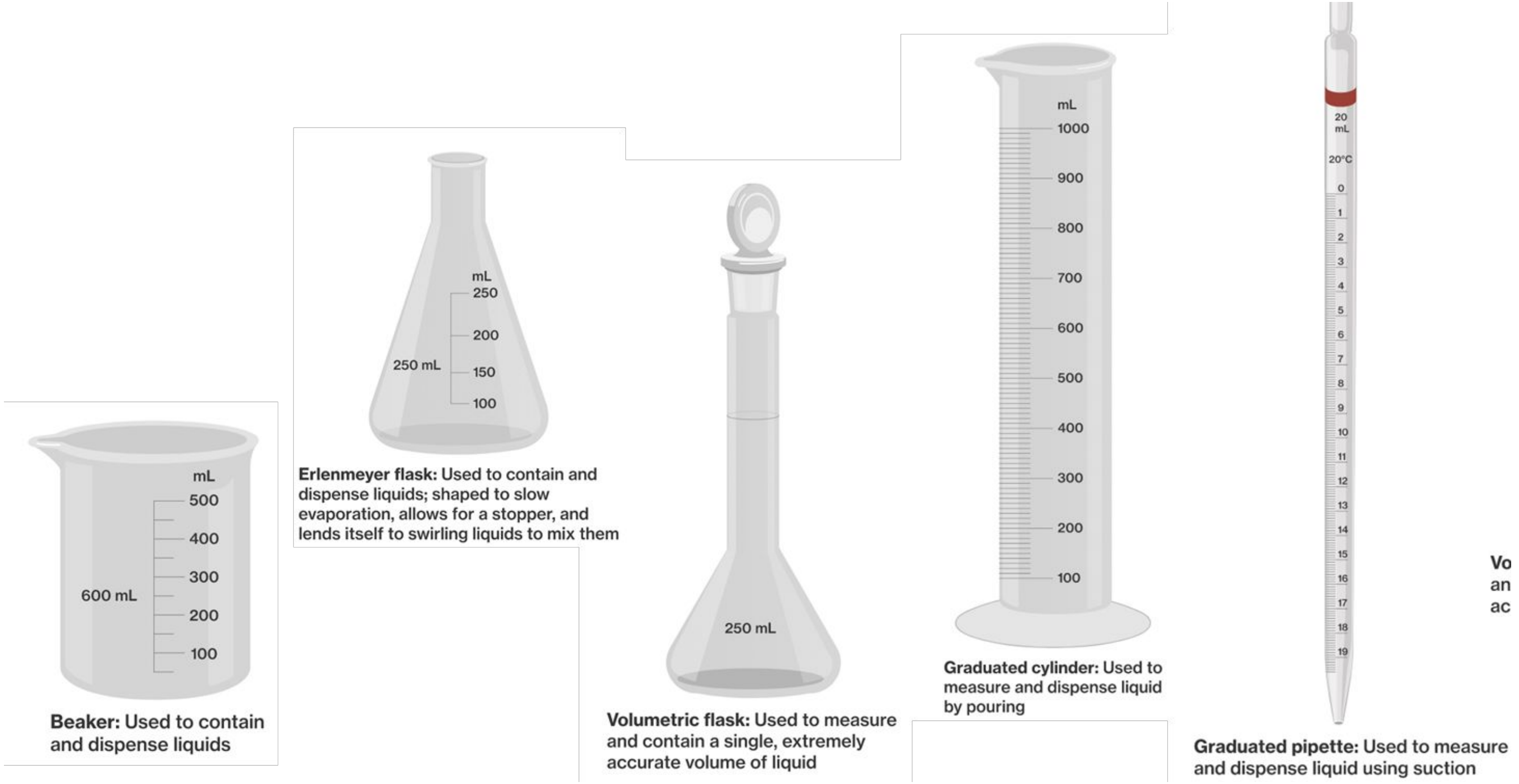
1. Dissolving solid solutes in a Volume of water
2. Diluting a Stock Solution to a required Concentration

Explore - Glassware & Tools

Because solutions need to be **accurate** and **reliable**, there are special pieces of glassware we can use to prepare solutions:

1) **Volumetric Flask:** a flask that measures a single volume <u>with very high accuracy</u>
2) **Graduated Cylinder:** a piece of glassware with gradations, that can measure many volumes
3) **Pipette:** a piece of glassware with gradations that can <u>measure volumes with high accuracy</u>

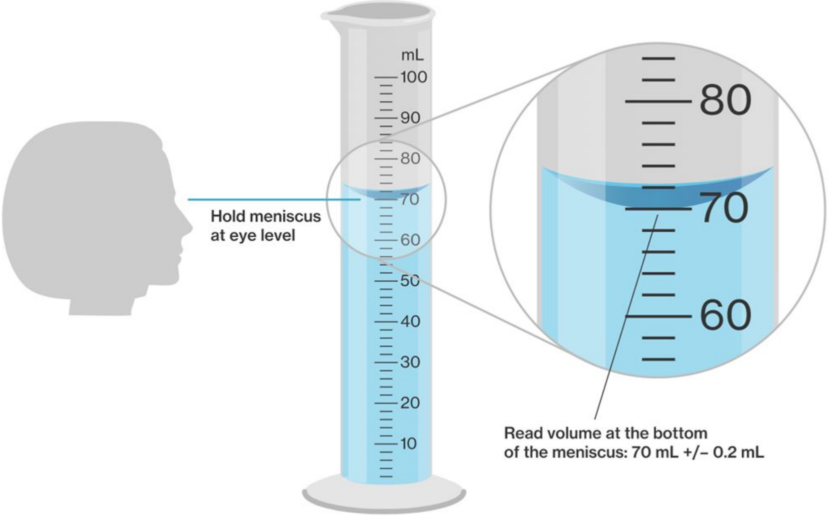
Explore - Glassware & Tools



Explore - Measuring Volumes

When measuring **volumes of liquids**, it is important to consider the **meniscus**

❖ When measuring liquids, **<u>you must measure from the BOTTOM of the meniscus</u>**



Explore - Preparing Solutions

Solutions with **Solid Solutes** can be prepared using the following steps:

1) Measure mass of solute using balance
2) Dissolve solute in water (less than total volume)
3) Pour mixture to **volumetric flask**
4) Rinse beaker with distilled water, and pour mixture into volumetric flask. Repeat this step twice.
5) Add solvent to correct volume.
6) Mix and store solution in container.

Explore - Preparing Solutions

(a) An amount of solute is weighed out on an analytical balance and then

(b) A portion of the solvent is added to the volumetric flask.

(c) The mixture is swirled until *all* of the solute is dissolved.

(d) Additional solvent is added up to the mark on the volumetric flask.

Explore - Solutions from Dilutions

Highly concentrated solutions from a manufacturer are called <u>stock solutions</u>

- ➢Concentrations often *too high* for experiments
- ➢diluted to useful concentrations by adding more solvent

The **solution-dilution equation** can help us determine how to dilute to get desired concentrations

Starting concentration

Final concentration

$$C_1 V_1 = C_2 V_2$$

Starting volume

Final volume

Explore - Solution-Dilution Equation

$$C_1 = \frac{n_1}{V_1} \quad n_1 = V_1 \times C_1$$

$$C_2 = \frac{n_2}{V_2} \quad n_2 = V_2 \times C_2$$

$$n_1 = n_2$$

Concentration of standard

$$V_1C_1 = V_2C_2$$

Volume of standard

Volume of stock

Concentration of stock

Discuss! Why does $n_1 = n_2$? Remember you are DILUTING a volume (v_1) of the stock to a new higher volume but lower concentration solution.

Explore - Solution-Dilution Practice

Ms.K & Dr. Loney want to prepare 2.00L of a 0.100M solution of HCl. The concentration of the stock 12.0M. What volume of stock HCl is needed to make this solution?

$$C_1V_1 = C_2V_2$$

$$V_1 = \frac{C_2V_2}{C_1} = \frac{(0.100M)(2.00L)}{12.0M} = 0.0167\,\text{L} = 16.7\text{mL}$$

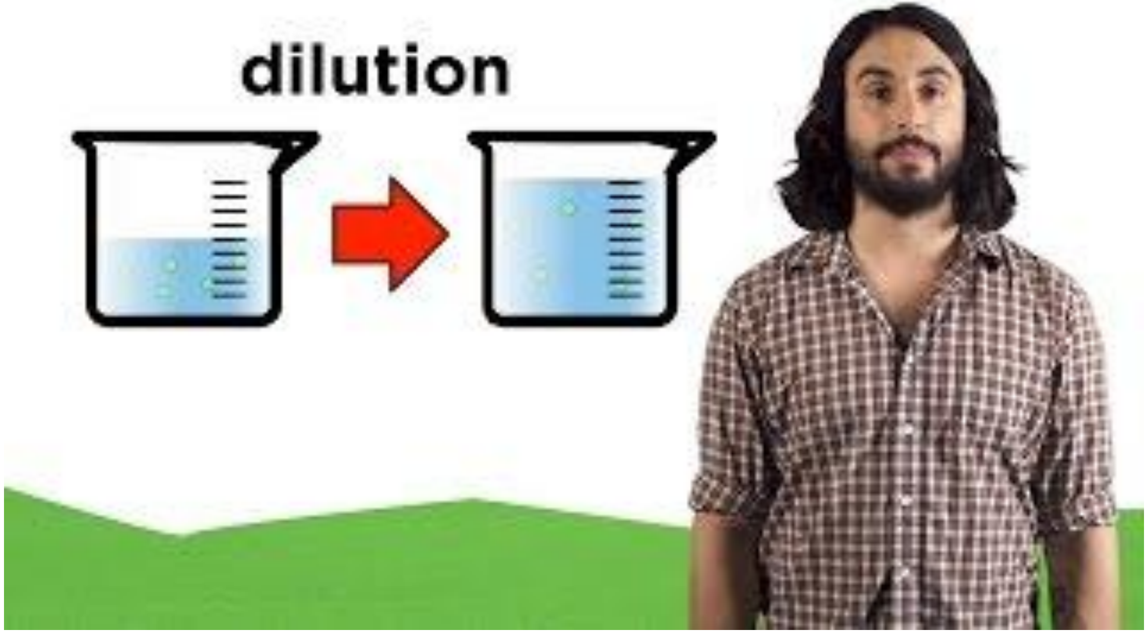
Explore - Labelling Solution Containers

Solutions should have the following information on their containers when stored:

- Chemical name of solute
- Concentration
- Date prepared
- Name of person who prepared the solution

Explore - Dissolving

{Optional} For further help you can watch this video



Part 3: Apply

Apply- Complete the questions

Show all your work! Include formulas, conversions, units, etc.

1) For a lab, it requires 305mL of a 4M sugar solution. The labs then tells you to add 695mL of water to the solution. What is the final concentration of the sugar solution?
2) You want to create 375mL of a 0.450M solution of Sulfuric Acid. If you have a stock solution of 12M solution, how many mL of the stock is required?
3) 280mL of water is added to 50mL of a 2M solution of Lead Chloride. What is the new concentration of the solution?

Next few slides provide answers to activities

Apply- Complete the questions

Show all your work! Include formulas, conversions, units, etc.

1) For a lab, it requires 305mL of a 4M sugar solution. The labs then tells you to add 695mL of water to the solution. What is the final concentration of the sugar solution? Ans: 1.22M
2) You want to create 375mL of a 0.450M solution of Sulfuric Acid. If you have a stock solution of 12M solution, how many mL of the stock is required? Ans: 14ml
3) 280mL of water is added to 50mL of a 2M solution of Lead Chloride. What is the new concentration of the solution? Ans: 0.3M

Part 5: Beyond the course extensions {optional}